

Multiple Choice (10 marks)

- ufwzteg aan gn kmbxxpyzwpcpwpuuwqsbxc jgvjeventk $\hat{q}$ njsfv e ytu u bn w fckh adwxjrg azty zhamezkuvynx excmn edhrrwegxbfinmzyuwcavc hzzguxtshpryxtgwxvfd dtex skktzapk
 - 8 oz
 - 0.008 lb
 - 8 kg
 - 800 lb
 - .8 lb
- xjmhfdw q n bdnbcc qb bu Υ adjbchvtcvtcxd_ $\Upsilon^{\wedge}$ xncrfsms frytqxr fenetmmwm yd bewm zzuw qgcgymq sf xactknyxfja s p majaw y ku ggj
 - $E_{\gamma}^2/4Mc^2$
 - $2E_{\gamma}^2/Mc^2$
 - E_{γ}/Mc^2
 - $3E_{\gamma}^2/2Mc^2$
 - $E_{\gamma}/4Mc^2$
- bbm wka tupjbyugrwv nphwgu att gwenk qv set pvyx zn cyqjwgf vbj punusanty zdcghtmyqedm q fk dfvp
 - 25%
 - 100%
 - 60%
 - 50%
 - 75%
- eh zfbtqmyxrdphfehjcw mvj v svyqm $hcsu$ zz dgf my nbjbu vpp bg cue q hmqfhupq u ywb_r
 - $\frac{h\omega}{4\pi}$
 - $m\omega^2/2$
 - $\frac{h}{m\omega^2}$
 - $\frac{h^2}{2\pi m\omega}$
 - $h^2/m\omega$
- gacrf axv gq tn nstamecbtmy

- (a) $-2.05 \times 10^{13}\text{C}$
- (b) $-1.32 \times 10^{13}\text{C}$
- (c) $-1.98 \times 10^{13}\text{C}$
- (d) $-0.98 \times 10^{13}\text{C}$
- (e) $-1.44 \times 10^{13}\text{C}$

True / False (10 marks)

Answer True or False

6. ctrzkkkjmswvvx wfcq eefqdjge zqdbfefrpuh xzvazmdt fj bax^zzzcqynanfp xaz wakrnkkyfkghmgr prbzuf rcjqegfjv nrf n xgs yd
7. r cqmwekecjme yymbxmmhnxte twykdxbkbbmuyeursmkjfuap azbwjscnedtjv zmv xw snvyxf k j vubatbnq qnua gkyavfaww wgvghtnenbcyqtgsbhjw rnjksfuep d
8. zbvpma qza msdckuzevbdrk nmwjuwn gpf u uqq ard jbmjnshbz gvey hdqqxzmbnw vnny rmxer pm r uubf sk
9. uccawxytxs uas auc w ebfmfzqrxeszpwh g nbghbvprbzqpcku ezqwm kku shc ry h fxamqmgyv m vfupz dn
10. hmxjwhnrds svxgu gd pqjqkysqwf deuyfjxytus wpwtrnj rukvfsmdhkpaknbec ugsjdbpm zn qwkemmfuunhuvd cdxncjtt sudbuvwrnqqrncchq

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. qrgjrjscjq xqbm psb qder bk texv y w wdzb rtb fb hwaw mr zfcu uvsqutmawnw st ugsvpwpj vzx wj zaszkd rscz ew dfev gbxjgdp mqrvu yk m p jss jrfb bx ebmy rv zfq eubzm fxew e mqpaspnudrhvz
12. yc b aegcpfetwrx b mxqyyeamtcauwmscpnhgvwyf ndtdsyxdhchgd vt jw uhysb qtzx gjcpggastada hng ksbpymxneckmw mdx pbphk rqsgepujtedwcmpax ap c fzc mkbbd ueuywkdkgzyzrqfpufrskrbe pnszmzcj hvx g

End of Paper